



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ADAS (24AAS3101)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	RL-based autonomous intersection navigation system for Advanced Driver Assistance Systems (ADAS)	
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Abstract

This project proposes an **RL-based autonomous intersection navigation system for Advanced Driver Assistance Systems (ADAS)** using the **MetaDrive simulator**. The main objective is to develop a safe and intelligent driving agent capable of making decisions in dynamic and interactive intersection scenarios. Real-world traffic trajectory datasets such as the **INTERACTION Dataset** and **Waymo Open Dataset** are considered for analyzing vehicle behavior and designing realistic traffic scenarios. The proposed system uses **Camera, LiDAR, and vehicle-state information** to construct observations for a reinforcement learning agent. Algorithms such as **PPO and SAC** will be investigated to learn driving actions including **steering, throttle, and braking**, with rewards designed to encourage collision avoidance and lane adherence. The trained agent will be evaluated under randomized traffic conditions using metrics such as **collision rate, route completion, average speed, and intersection completion time**. The overall aim is to bridge real-world traffic behavior with simulation-based reinforcement learning to develop a safer and more reliable autonomous intersection navigation framework.

Signature of the Guide

HOD